

1 The Likelihood Ratio as a Sufficient Statistic

1.1 Definition

For each sensor i , define the likelihood ratio of its observation as

$$l^i := L(y^i) := \frac{f_{Y^i|H}(y^i | H_2)}{f_{Y^i|H}(y^i | H_1)} \quad a.s. \quad (1)$$

a single real number capturing how much more consistent the raw observation y^i is with H_2 than with H_1 .

1.2 Why l^i is sufficient

l^i is a sufficient statistic for H : once its value is known, nothing else about the raw observation y^i can further move one's belief about which hypothesis is true. With only two hypotheses on the table, the entire way in which the observation's density can vary with H is exhausted by a single comparison - how large the density is under H_2 relative to how large it is under H_1 , at that particular observation. A single number is therefore inherently enough to hold everything that changes with H , and that number is, by construction, exactly l^i . (How this shows up concretely inside the paper's own machinery - via Lemma 1 - is worth treating separately once we get there.)

1.3 No information loss: a Markov-chain and DPI reading

Sufficiency means that any two raw observations producing the same l^i are, for the purpose of distinguishing H_1 from H_2 , interchangeable: whatever detail of y^i is discarded in forming l^i carries no evidential weight about H . This has a clean information-theoretic statement. Since l^i is a deterministic function of y^i , the chain

$$H \rightarrow y^i \rightarrow l^i$$

is trivially Markov, and the Data Processing Inequality (DPI) gives

$$I(H; l^i) \leq I(H; y^i). \quad (2)$$

Equality holds *iff* l^i is sufficient for H - equivalently, *iff* $H \rightarrow l^i \rightarrow y^i$ is *also* a Markov chain, i.e. y^i carries no further information about H once l^i is fixed, gives equality

$$I(H; l^i) = I(H; y^i). \quad (3)$$